Mediation Analysis in Production Functions with Unobserved Productivity

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Abstract

Economists are often interested in knowing the causal mechanism by which a treatment variable impacts a particular outcome variable, whether directly, indirectly, or both. This question is not limited to reduced-form settings; it also includes production settings where researchers want to know whether a treatment variable affects output directly, indirectly, or both. Decomposing the total effect of a treatment into direct and input-mediated channels is a fundamental challenge in production economics. Standard mediation analysis is generally invalid in these settings because firms choose inputs in response to productivity observed by the firm but hidden from the econometrician. We show how this endogenous mediator choice biases the conventional product-of-coefficients decomposition and develop a structural framework to consistently identify direct, indirect, and total treatment effects. Monte Carlo experiments across both additive and treatment–mediator interaction models demonstrate that our feasible ACF-style estimator accurately recovers the component effects. Conversely, deliberately omitting productivity from the mediator equation introduces substantial upward bias in the treatment–mediator coefficient and the resulting indirect effect. We apply this method to 13,902 observations from 2,416 Spanish manufacturing firms, treating labor as the mediator and a one-year-lagged digitalization index as the treatment.

Keywords: direct and indirect effects; proxy variables; generated regressors; generalized method of moments; firm-level panel data; digital adoption

JEL classification: C31, C33, D24, O33

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1 Introduction

Economists are frequently interested in decomposing the total effect of a treatment variable on an outcome into direct and indirect channels. In a production-function setting, this distinction is particularly important. A treatment, such as firm investment in digital technologies, may increase output directly by improving the underlying production technology, while simultaneously affecting output indirectly by altering the firm's optimal demand for conventional inputs like labor or capital. In this context, the affected input serves as a mediator. Quantifying this indirect channel requires identifying two distinct parameters: the effect of the treatment on the mediator, and the effect of the mediator on output. Consequently, the indirect effect is characterized by the product of a parameter from the mediator equation and an elasticity from the production function.

Identifying these two components presents a unique econometric challenge due to the pervasive issue of unobserved productivity. While structural production-function estimators can successfully identify the mediator's output elasticity and the treatment's direct effect under appropriate conditions, they do not inherently identify the treatment's effect on the mediator. This latter parameter must be recovered from the firm's input demand equation. However, firms with higher unobserved productivity may simultaneously adopt more digital technologies and adjust their use of labor. If this unobserved productivity is not explicitly controlled for in the mediator equation, the estimated effect of the treatment on the mediator will be biased. Because this parameter enters the indirect effect multiplicatively, this bias fundamentally contaminates the causal mediation analysis.

In this study, we consider a setting in which a treatment enters the production function directly and also operates indirectly by changing the firm's use of a conventional input. Specifically, we formalize a causal mediation framework in which labor mediates the relationship between digitalization and output. The total effect of the treatment is thus decomposed into a direct output effect and an indirect effect operating through labor allocation. To our knowledge, this study is the first to develop an econometric framework for causal mediation strictly within production-function estimation, where unobserved productivity acts as a common confounder throughout treatment, mediator, and outcome.

We make three primary methodological contributions. First, we formally show that correcting only the production function for endogenous input choice is insufficient for mediation analysis; recovered productivity must also enter the mediator equation when it confounds treatment and mediated input. Second, we develop a sequential estimator that integrates an ACF-style generalized method of moments (GMM) approach (Ackerberg et al., 2015) with a productivity-controlled mediator regression. To account for the uncertainty introduced by the generated productivity control, we propagate the first-stage error through a firm-block bootstrap that repeats every estimation stage. Third, we derive the direct, labor-mediated, and total effects under specifications both with and without a treatment–labor interaction.

We validate our estimator using Monte Carlo simulations. We specify a data-generating process

(DGP) that mirrors realistic empirical settings, where conventional inputs, the mediator, and the treatment variable are all endogenously chosen conditional on unobserved productivity. The ACF-style GMM procedure successfully recovers the production-function parameters and the treatment effect in the first stage, while simultaneously extracting a consistent estimate of the latent productivity state. By including this recovered productivity as a control function in the mediator equation, we obtain unbiased estimates of the treatment's effect on the mediator. Our simulations demonstrate that failing to control for unobserved productivity in the mediator equation induces substantial omitted-variable bias in the estimated indirect effect, whereas our proposed structural correction reduces this bias to near zero.

We then apply our methodology to Spanish manufacturing panel data to examine the impact of digitalization on value-added output. In this empirical application, the treatment variable is a comprehensive digitalization index, the mediator is labor hours, and the outcome is value added. In the baseline Cobb-Douglas production model—where there is no interaction between the treatment and the mediator—our productivity-controlled estimates are statistically significant, with confidence intervals that strictly exclude zero. However, when we estimate a flexible specification that includes a treatment–labor interaction, the estimated interaction parameter is statistically indistinguishable from zero. Finally, we quantify the magnitude of the omitted-variable bias by comparing our fully controlled estimates to a naive specification. Consistent with our Monte Carlo evidence, the empirical results confirm that failing to control for latent productivity in the mediator equation severely upward-biases the estimated indirect effect.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 sets up the model and defines the estimands. Section 4 presents the Monte Carlo simulations. Section 5 describes the empirical application, and Section 6 discusses the results. Finally, Section 7 summarizes and concludes.

2 Related Literature

Causal mediation analysis decomposes a treatment effect into a component that operates through a specified mediator and a component that operates through other pathways (Robins and Greenland, 1992; Imai et al., 2010; VanderWeele, 2015; Celli, 2022). In a linear model without interactions, the familiar product-of-coefficients indirect effect multiplies the treatment–mediator coefficient by the mediator–outcome coefficient. Its causal interpretation nevertheless requires restrictions on both treatment assignment and mediator assignment. In particular, sequential ignorability rules out unobserved variables that affect the mediator and the outcome after conditioning on treatment and observed covariates (Imai et al., 2010).

Several econometric approaches relax parts of this requirement. Instrumental variable methods identify direct and indirect effects under additional exclusion restrictions (Frölich and Huber, 2017), while the average controlled direct-effect procedure of Acharya et al. (2016) targets mechanisms without con-

ditioning naively on a post-treatment variable. In a production-technology setting, Heckman and Pinto (2015) show how experimentally estimated technologies can be used to identify sources of treatment effects when inputs are unmeasured or measured with error. However, applying these mediation methods to observational production settings remains challenging because firms choose their inputs after observing productivity components that remain hidden from the econometrician. We cannot rely on standard experimental methods because this unobserved productivity confounds the inputs, the mediated variables, and the treatment, leaving both the treatment and the mediated input correlated with the error term.

The structural production-function literature addresses precisely this simultaneity problem. Firms choose capital, labor, and flexible inputs in response to productivity observed by the firm but not by the econometrician. Proxy-variable estimators use investment or intermediate-input demand to invert the firm's input policy and control for the latent productivity state (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Ackerberg et al., 2015). Related work allows innovation or other firm decisions to shift the productivity process and distinguishes Hicks-neutral from factor-biased productivity (Doraszelski and Jaumandreu, 2013; Wang et al., 2025).

While these estimators successfully identify production elasticities and recover a productivity control, they do not automatically identify a causal mediation path. If treatment changes labor, the indirect effect also depends on the treatment coefficient in the labor equation. When productivity affects both treatment and labor, correcting only the production function leaves this second coefficient confounded. Identifying the fully mediated path requires utilizing the recovered productivity twice, first to identify the production coefficients, and then as a control in the input equation.

A closely related literature attempts to bridge this gap by estimating productivity and subsequently studying mechanisms involving that estimated object. For instance, Chemo Dzukou and Vancauteren (2024) examine whether innovation inputs and outputs mediate the causal effect of export-market entry on productivity in the French dairy industry. They achieve this by embedding an ACF-style control function within an extended system estimated by simulated maximum likelihood. However, their framework evaluates productivity as the ultimate outcome, rather than analyzing a structural setting where a treatment variable enters the production function directly and indirectly affects output through conventional inputs.

Similarly, Janes et al. (2026) study randomized capital grants to microenterprises by recovering TFP and applying the average controlled direct-effect method of Acharya et al. (2016). Because their treatment is randomly assigned, their comparison is fundamentally different from an observational setting where both the treatment and an input mediator respond endogenously to latent productivity. Other applications insert estimated proxy-TFP into conventional mechanism regressions, treating it either as a mediator (Qiu et al., 2024; Wang et al., 2022) or an outcome (Fan et al., 2026). While these papers provide substantive evidence regarding various productivity mechanisms, their estimands fundamentally differ from an input-mediated output effect that is strictly contained within a structural

production function.

The novelty here is therefore not the use of multiple estimation stages, but rather the structural linkage of the stages for a different causal object. Output is the outcome, labor is simultaneously a chosen production input and the mediator, and latent productivity is a common cause of treatment, labor, and output. By combining the production elasticity with the productivity-controlled treatment–labor coefficient, and permitting treatment to change the labor elasticity, this formulation shows why a consistent estimate of the production function alone does not guarantee a consistent indirect effect.

3 Model

3.1 Technology and the labor-mediation system

Let's consider firm i in period t that produces output using capital and labor according to the following Cobb-Douglas technology:

$$y_{it} = \beta_0 + \alpha k_{it} + \beta l_{it} + \delta x_{it} + \omega_{it} + \varepsilon_{it}. \tag{1}$$

The Cobb–Douglas technology in (1) is in logarithmic form. The parameter α is the output elasticity of capital, β is the output elasticity of labor, and δ captures the direct effect of treatment on output, holding capital, labor, and productivity fixed. We consider a more flexible specification in which treatment interacts with labor and, consequently, changes the output elasticity of labor.

The term ω_{it} denotes Hicks-neutral productivity that is observed by the firm when it makes its input decisions but is unobserved by the econometrician. The term ε_{it} is an ex-post output shock, including measurement error, satisfying $E[\varepsilon_{it} \mid \mathcal{I}_{it}] = 0$, where $\mathcal{I}_{it}$ denotes the firm's information set at time t. Productivity follows a first-order Markov process:

$$\omega_{it} = g(\omega_{i,t-1}) + \xi_{it}, \qquad E[\xi_{it} \mid \mathcal{I}_{i,t-1}] = 0. \tag{2}$$

In both specifications, labor serves as a mediator through which treatment affects output. The labor-mediator equation is

$$l_{it} = a_0 + a_1 x_{it} + \phi \omega_{it} + z_{it}'\pi + v_{it}, \qquad E[v_{it} \mid x_{it}, \omega_{it}, z_{it}] = 0, \tag{3}$$

where a_1 measures the effect of treatment on labor and ϕ captures the response of labor to firm productivity. The vector z_{it} contains other observed determinants of labor, excluding the treatment x_{it}.

When treatment is not randomly assigned, treatment adoption may be correlated with firm productivity $\mathrm{Cov}(x_{it}, \omega_{it}) \neq 0$. For example, more productive firms may be more likely to adopt digital technologies and may simultaneously adjust their employment. Consequently, omitting productivity from the labor equation generally confounds the estimated effect of digitalization on labor.

Even if treatment is randomly assigned, productivity may continue to confound the relationship between labor and output because productivity affects both the firm's labor choice and its output.

Thus, random assignment of treatment alone does not eliminate mediator–outcome confounding.

3.2 Case 1: Baseline model without treatment–labor interaction

We first consider the baseline production function in (1), in which treatment and labor enter additively. Because treatment and labor do not interact, the output elasticity of labor is constant β.

According to the labor equation (3), a one-unit increase in treatment changes labor by a_1. A one-unit increase in labor, in turn, changes output by β. Therefore, for a one-unit change in treatment, the direct, labor-mediated, and total effects are

$$DE = \delta, \qquad IE_L = a_1\beta, \qquad TE = \delta + a_1\beta. \tag{4}$$

The parameter δ captures the direct effect of treatment on output, whereas $a_1\beta$ captures the indirect effect operating through the treatment-induced change in labor.

To obtain the corresponding reduced form, substitute the labor equation (3) into the production function (1):

$$\begin{aligned} y_{it} &= \beta_0 + \alpha k_{it} + \beta\left(a_0 + a_1 x_{it} + \phi\omega_{it} + z_{it}'\pi + v_{it}\right) + \delta x_{it} + \omega_{it} + \varepsilon_{it} \\ &= (\beta_0 + \beta a_0) + \alpha k_{it} + (\delta + a_1\beta)x_{it} + (1 + \beta\phi)\omega_{it} + \beta z_{it}'\pi + \beta v_{it} + \varepsilon_{it}. \end{aligned} \tag{5}$$

The reduced-form coefficient on treatment is $\delta + a_1\beta$, which represents the total effect of treatment on output. Estimating the reduced-form equation alone therefore does not separately identify the direct effect δ and the indirect effect $a_1\beta$. Identification of the indirect effect requires combining an estimate of the labor elasticity β from the structural production function with an estimate of a_1 from the labor-mediator equation.

The ACF procedure can be used to address the endogeneity of labor and recover the structural production-function parameters, including β and δ, under the appropriate identification assumptions. It does not, however, identify a_1, because this parameter belongs to the labor-mediator equation. The labor-mediator equation must therefore be estimated separately while controlling for the productivity term recovered from the production- function estimation.

3.3 Case 2: Flexible model with treatment–labor interaction

We next relax the constant-labor-elasticity restriction by allowing treatment to change the output elasticity of labor:

$$y_{it} = \beta_0 + \alpha k_{it} + \beta l_{it} + \delta x_{it} + \theta x_{it} l_{it} + \omega_{it} + \varepsilon_{it}. \tag{6}$$

This specification extends the baseline model by including an interaction between treatment and labor. The baseline model is recovered by imposing $\theta = 0$. The output elasticity of labor is now $\beta + \theta x_{it}$. Thus, β is the output elasticity of labor at $x_{it} = 0$, while θ measures how treatment changes this elasticity.

Holding labor fixed at l, the marginal effect of treatment on output is $\delta + \theta l$. Treatment therefore affects output through the baseline direct effect δ and through its interaction with the existing labor input, θl.

Because the output elasticity of labor depends on treatment, the labor-mediated effect also depends on the treatment level at which it is evaluated. For a continuous treatment, the labor equation implies a_1. Applying the chain rule, the marginal total effect of treatment on output is

$$\begin{aligned} \frac{dy_{it}}{dx_{it}} &= \frac{\partial y_{it}}{\partial x_{it}} + \frac{\partial y_{it}}{\partial l_{it}} \frac{\partial l_{it}}{\partial x_{it}} \\ &= \underbrace{\delta + \theta l_{it}}_{\text{direct effect holding labor fixed}} + \underbrace{a_1(\beta + \theta x_{it})}_{\text{indirect effect through labor}} . \end{aligned} \tag{7}$$

Hence, the marginal direct, indirect, and total effects in the flexible specification are

$$DE_{it} = \delta + \theta l_{it}, \qquad IE_{L,it} = a_1(\beta + \theta x_{it}), \qquad TE_{it} = \delta + \theta l_{it} + a_1(\beta + \theta x_{it}). \tag{8}$$

This decomposition separates two economically distinct mechanisms. The term θl_{it} captures how treatment changes the productivity of the existing labor input while holding the quantity of labor fixed. In contrast, $a_1(\beta + \theta x_{it})$ captures the effect arising because treatment changes the quantity of labor employed. Setting $\theta = 0$ reduces these expressions to the baseline model:

$$DE = \delta, \qquad IE_L = a_1\beta, \qquad TE = \delta + a_1\beta.$$

The flexible specification could be extended further by allowing the production-function coefficients to vary across firms using a random-coefficient model. Such an extension would accommodate broader forms of heterogeneity in the production technology. The present analysis, however, focuses on the baseline model without a treatment–labor interaction and the flexible model with the interaction specified in (6).

3.4 Identification

Consider a production setting where a firm invests heavily in digital technologies. This investment, often captured by a comprehensive digitalization index, encompasses human capital development, such as enhancing workers' proficiency with artificial intelligence (AI) and large language models (LLMs). When a firm makes these investment decisions, it already observes its latent productivity level. Consequently, both the firm's labor demand and its demand for digital technologies are endogenous; they are chosen conditional on productivity that remains unobserved by the econometrician.

A researcher evaluating this firm may wish to quantify the output gains from these digital investments, recognizing that the digital technologies do not merely shift the production frontier directly; it also exerts an indirect effect on output by altering the firm's optimal labor allocation. Under conventional causal mediation frameworks, decomposing the total effect of digital technology into these direct and indirect channels is generally intractable. The unobserved productivity acts as a pervasive confounder, simultaneously contaminating the treatment (digital technology), the mediator (labor),

and the conventional inputs. To overcome this simultaneity problem, we propose a structural method that controls for this latent productivity, allowing for a consistent decomposition of the direct and input-mediated effects on output.

To formalize this, a causal interpretation of these estimands requires sequential ignorability assumptions that address confounding in both the treatment–output and labor–output relationships. Specifically, conditional on observed covariates and the latent productivity state, the treatment must be independent of the relevant potential labor and output outcomes. Furthermore, the mediator must be independent of the potential output outcomes conditional on treatment, observed covariates, and productivity.

Unobserved productivity fundamentally threatens these conditions because it simultaneously dictates treatment adoption, labor demand, and output. Even in a hypothetical scenario where x_{it} were experimentally assigned, productivity would continue to confound the labor–output relationship whenever it enters the labor demand equation. Omitting this productivity control from the labor equation would therefore severely bias the treatment–labor coefficient, a_1, and, consequently, the estimated indirect effect.

In the baseline additive model, identifying the labor-mediated effect requires consistent estimates of the labor elasticity (β) from the production function and the treatment effect on labor (a_1) from the mediator equation. Crucially, the production function alone cannot identify a_1. In a more flexible model that allows the treatment to alter the labor elasticity, identification additionally requires a consistent estimate of the interaction parameter (θ). Because labor (l_{it}), treatment (x_{it}), and their interaction ($x_{it}l_{it}$) are all correlated with unobserved productivity, the production-function estimator must explicitly address the endogeneity of all three terms. Accordingly, the interaction model generally requires additional moment conditions or instruments within the GMM framework. Once β, θ, and a_1 are consistently identified, the marginal or natural indirect effects can be accurately evaluated at the relevant treatment levels.

4 Simulations

In this section, we perform Monte Carlo simulations to see how the estimator behaves when we control for unobserved productivity in the mediator equation and when we do not. Our simulation, through our data-generating process, shows that estimates of the indirect effects are biased when the treatment effect is confounded with unobserved productivity; controlling for unobserved productivity in the mediator makes the estimates unbiased.

To perform these simulations, we examine the two production functions in Section 3 using only the feasible ACF proxy-variable estimator. Each design has $N = 500$ firms, $T = 6$ observed periods, and 500 Monte Carlo replications. The common parameters are $\alpha = 0.35$, $\beta = 0.60$, $\delta = 0.10$, $a_0 = 0.20$, $a_1 = 0.50$, and $\phi = 0.60$. Productivity follows $\omega_{it} = 0.70\omega_{i,t-1} + \xi_{it}$, with $\xi_{it} \sim N(0, 0.40^2)$. Fifty periods

are discarded before each panel is retained, so the results do not depend on arbitrary initial conditions.

To examine a situation when the treatment is confounded with unobserved productivity, we make the treatment to be persistent and responds to current productivity,

$$x_{it} = 0.50x_{i,t-1} + 0.35\omega_{it} + e^x_{it},$$

whereas capital is predetermined relative to the current productivity innovation,

$$k_{it} = 0.65k_{i,t-1} + 0.40\omega_{i,t-1} + e^k_{it}.$$

Labor is generated from equation (3), and output is generated from equation (1) in Case 1 and equation (6) in Case 2. The standard deviations of e^x_{it}, e^k_{it}, v_{it}, and ε_{it} are, respectively, 0.70, 0.70, 0.80, and 0.30.

4.1 Estimation

The simulations implement the ACF proxy-variable approach to identify the production-function parameters. The proxy is the observed material demand, denoted as q_{it}. The simulated output measure is value added, so materials do not enter the production function as a separate productive input. Instead, materials are a flexible input chosen after the firm observes current productivity: $q_{it} = 0.15 + 0.20k_{it} + 0.15l_{it} + 0.10x_{it} + \omega_{it} + 0.15\tanh(\omega_{it})$. Conditional on (k_{it}, l_{it}, x_{it}), this demand function is strictly increasing in productivity because $\frac{\partial q_{it}}{\partial \omega_{it}} = 1 + 0.15\,\mathrm{sech}^2(\omega_{it}) > 0$. Thus, materials demand is informative about the scalar productivity state and can be inverted.

ACF estimation of productivity follows a two-stage approach. The first stage regresses output on a quadratic sieve in $(k_{it}, l_{it}, x_{it}, q_{it})$ augmented by q^3_{it}, producing the fitted conditional mean $\widehat{\Phi}_{it}$. For a candidate production-parameter vector b, recovered productivity is $\widehat{\omega}_{it}(b) = \widehat{\Phi}_{it} - r'_{it}b$, where $r_{it} = (k_{it}, l_{it}, x_{it})'$ in Case 1 and $r_{it} = (k_{it}, l_{it}, x_{it}, x_{it}l_{it})'$ in Case 2. At every candidate, we fit a linear first-order Markov law to recovered productivity and form its innovation. Case 1 uses current capital, lagged labor, and lagged treatment as instruments. Case 2 adds the lagged treatment–labor interaction. Finally, we estimate equation (3) by OLS after including the recovered productivity control.

4.2 Case 1: no treatment–labor interaction

The ACF estimates are centered close to the data-generating values. The mean labor elasticity is 0.600 against a truth of 0.600, and the mean treatment coefficient is 0.096 against a truth of 0.100. Combining the production estimate with the productivity-controlled labor equation gives a mean indirect effect of 0.299 against 0.300 and a mean total effect of 0.395 against 0.400. The mediated-share mean is slightly above its population value because it is a nonlinear ratio; the component estimates themselves show little bias.

Table 1: Case 1: ACF estimates without a treatment–labor interaction

Estimand	True value	Mean	MC SD	RMSE
α	0.350	0.350	0.018	0.018
β	0.600	0.600	0.016	0.016
δ	0.100	0.096	0.052	0.053
a_1	0.500	0.498	0.037	0.037
DE	0.100	0.096	0.052	0.053
IE_L	0.300	0.299	0.025	0.025
TE	0.400	0.395	0.072	0.072
IE_L/TE	0.750	0.774	0.101	0.103

Notes. Feasible ACF proxy-variable estimator; $N = 500$ firms, $T = 6$ periods, 500 replications (500 converged). The first stage uses a quadratic sieve in capital, labor, treatment, and materials demand, augmented by a cubic materials term. The second stage uses current capital and lagged labor and treatment as instruments. The labor equation includes the ACF productivity control. MC SD is the Monte Carlo standard deviation.

4.3 Case 2: treatment–labor interaction

Case 2 sets $\theta = 0.10$. To make the observation-specific effects in equation (8) comparable across replications, we evaluate them at $x = 1$ and at the labor level implied by the mediator equation at mean productivity and zero mediator shock, $l = a_0 + a_1 x = 0.70$. The population effects at this point are $DE = 0.10 + 0.10(0.70) = 0.17$, $IE_L = 0.50(0.60 + 0.10) = 0.35$, and $TE = 0.52$.

Table 2: Case 2: ACF estimates with a treatment–labor interaction

Estimand	True value	Mean	MC SD	RMSE
α	0.350	0.350	0.018	0.018
β	0.600	0.599	0.015	0.015
δ	0.100	0.100	0.053	0.053
θ	0.100	0.099	0.017	0.017
a_1	0.500	0.497	0.036	0.036
DE	0.170	0.169	0.057	0.057
IE_L	0.350	0.347	0.029	0.029
TE	0.520	0.517	0.080	0.080
IE_L/TE	0.673	0.681	0.065	0.065

Notes. Feasible ACF proxy-variable estimator; $N = 500$ firms, $T = 6$ periods, 500 replications (500 converged). The first stage uses a quadratic sieve in capital, labor, treatment, and materials demand, augmented by a cubic materials term. The second stage uses current capital and lagged labor and treatment as instruments, plus the lagged treatment–labor interaction. The labor equation includes the ACF productivity control. MC SD is the Monte Carlo standard deviation. Effects are evaluated at $x = 1$ and $l = a_0 + a_1 = 0.70$, as implied by the design at mean productivity.

The interaction coefficient is recovered with a mean of 0.100 and a Monte Carlo standard deviation of 0.017. The mean direct, indirect, and total effects are 0.169, 0.347, and 0.517, respectively, close to

their target values of 0.170, 0.350, and 0.520. The results show that the same ACF control-function logic accommodates the interaction once the instrument set also contains variation for $x_{it}l_{it}$.

4.4 Omitting productivity from the labor equation

We now examine a situation in which treatment and labor are both confounded with unobserved productivity, but productivity is omitted from the labor equation. We deliberately create the confounding described in Section 3.4: treatment loads on current productivity with coefficient 0.35, while labor loads on productivity with coefficient $\phi = 0.60$. Across replications, the mean sample correlations are approximately 0.35 between treatment and productivity and 0.48 between labor and productivity. We retain the same ACF production estimates and change only the labor equation. The controlled specification includes recovered productivity; the misspecified equation omits it.

Table 3: Bias from omitting productivity from the labor equation

	True	Productivity controlled		Productivity omitted	
Estimand	value	Mean	Bias	Mean	Bias
Panel A: Case 1, additive					
a_1	0.500	0.498	-0.002	0.634	0.134
IE_L	0.300	0.299	-0.001	0.381	0.081
TE	0.400	0.395	-0.005	0.476	0.076
IE_L/TE	0.750	0.774	0.024	0.808	0.058
Panel B: Case 2, interaction					
a_1	0.500	0.497	-0.003	0.631	0.131
IE_L	0.350	0.347	-0.003	0.441	0.091
TE	0.520	0.517	-0.003	0.623	0.103
IE_L/TE	0.673	0.681	0.008	0.712	0.039

Notes. The DGP makes treatment and labor depend on current productivity: the treatment loading is 0.35 and the labor-equation loading is $\phi = 0.60$. Both columns use the same feasible ACF production estimates. The controlled labor equation includes recovered productivity; the omitted equation regresses labor only on treatment and an intercept. Bias is the Monte Carlo mean minus the true value. $N = 500$, $T = 6$, 500 replications; converged replications are 500 in Case 1 and 500 in Case 2. Case-2 effects are evaluated at $x = 1$.

Table 3 provides the intended bias check. With the productivity control, the mean bias in a_1 is between -0.003 and -0.002 across the two cases. Omitting productivity raises the mean estimate from about 0.50 to about 0.63, producing upward biases of 0.134 in Case 1 and 0.131 in Case 2. The indirect-effect bias rises from essentially zero to 0.081 and 0.091, respectively, and the total effect is also overstated. Thus, correcting the production function alone is insufficient: productivity must also be controlled in the labor equation when it confounds treatment and labor.

5 Application to Digitalization and Labor

We apply our method to the Spanish manufacturing industry where we examine the direct and indirect effects of digital technologies on output. The mediator is labor, measured as log effective labor hours. We use firm-level data from the Encuesta Sobre Estrategias Empresariales (ESEE), administered by Spain's Ministry of Industry, for 2000–2019. The ESEE reports firm-specific output and intermediate-input price changes, which we use to construct real production and real intermediate consumption following the general deflation approach in Jaumandreu and Lin (2018). Real value added is real production minus real intermediate consumption. Capital is the real capital stock, labor is total effective hours, and real intermediate consumption is the flexible-input proxy.

The treatment is the one-year lag of the digitalization index. The index lies on $[0, 1]$ and combines thirteen indicators of IT technology, digital human capital, automation, and stakeholder interaction, following Añón Higón and Bonvin (2024). We form the lag only when the firm is observed in consecutive years. Labor is the mediator, so the empirical chain is lagged digitalization $\rightarrow$ labor hours $\rightarrow$ value added.

We require positive value added, capital, labor, and materials. We then trim the logs of these four variables at the 2.5th and 97.5th percentiles; the bounded treatment is not trimmed. The proxy first stage contains 17,277 firm-year observations on 2,762 firms. Requiring the consecutive lags needed for the ACF productivity innovation leaves 13,902 observations on 2,416 firms. The estimation sample covers 2001–2019.

5.1 Specification and estimation

We estimate both production functions in Section 3. Case 1 is the additive model in (1). Case 2 adds the treatment–labor interaction in (6). Output, capital, labor, and materials enter in natural logarithms; lagged digitalization enters in levels. Thus, a_1 is the change in log labor hours associated with a one-point change in the digitalization index.

For each year, the ACF first stage uses a quadratic sieve in capital, labor, treatment, and materials, augmented by a cubic materials term. At every candidate production-parameter vector, the productivity law is refitted as a year-specific cubic in lagged recovered productivity. Case 1 uses current capital, lagged labor, and current treatment as its three moments. Current treatment is already lagged digitalization and is therefore predetermined relative to the current productivity innovation. Case 2 adds an instrument equal to current treatment times lagged labor for the treatment–labor interaction. Both structural systems are exactly identified, so a Hansen overidentification test is not available.

The labor equation includes lagged digitalization, recovered current productivity, log capital, and year indicators. Log capital is included as an observed determinant in z_{it} because employment scale depends directly on the firm's capital stock. Case 2 is estimated exactly as written in (6), using the uncentered variables l_{it}, x_{it}, and $x_{it}l_{it}$. Thus, β is the labor elasticity at $x = 0$ and δ is the treatment

coefficient at $l = 0$. Because the marginal effects vary with treatment and labor, the reported Case-2 indirect and direct effects are sample averages, numerically equal to evaluating them at $\bar{x}$ and $\bar{l}$, respectively. Uncertainty is measured with 999 firm-block bootstrap draws that repeat the proxy first stage, productivity law, structural GMM, and labor equation.

5.2 Results

Table 4: Labor mediation of the effect of lagged digitalization on value added

	Case 1: additive		Case 2: interaction	
Estimand	Estimate (SE)	95% interval	Estimate (SE)	95% interval
Capital elasticity α	0.200 (0.022)	[0.188, 0.272]	0.201 (0.022)	[0.189, 0.274]
Labor coefficient β	0.885 (0.030)	[0.789, 0.908]	0.869 (0.039)	[0.758, 0.913]
Treatment coefficient δ	0.220 (0.061)	[0.082, 0.327]	-0.180 (0.555)	[-1.305, 0.831]
Treatment–labor interaction θ	–	–	0.034 (0.048)	[-0.056, 0.130]
Treatment effect on labor a_1	0.524 (0.066)	[0.436, 0.684]	0.525 (0.066)	[0.436, 0.685]
Indirect effect IE_L	0.464 (0.058)	[0.361, 0.592]	0.463 (0.058)	[0.360, 0.589]
Direct effect DE	0.220 (0.061)	[0.082, 0.327]	0.217 (0.061)	[0.081, 0.319]
Total effect TE	0.684 (0.079)	[0.520, 0.826]	0.680 (0.079)	[0.520, 0.823]
Mediated share IE_L/TE	0.679 (0.074)	[0.572, 0.866]	0.680 (0.074)	[0.575, 0.868]

Notes. Output is log real value added; labor is log effective hours; treatment is the consecutive one-year lag of the digitalization index; and log real materials is the ACF proxy. The labor equation controls for recovered productivity, log capital, and year indicators. Case 1 has a constant labor elasticity and direct treatment effect. Case 2 uses uncentered treatment, labor, and their interaction. Its reported IE_L and DE are sample-average marginal effects, evaluated at mean lagged digitalization and mean labor, respectively. Parentheses contain firm-block bootstrap standard errors (999 successful draws); intervals are bootstrap percentiles.

The two specifications give nearly the same average decomposition. In Case 1, the capital and labor coefficients are 0.200 and 0.885. The estimated direct effect of a full index-point increase in lagged digitalization is 0.220 log points. The controlled treatment coefficient in the labor equation is $\hat{a}_1 = 0.524$, giving an indirect effect of 0.464 and a total effect of 0.684. In the uncentered Case-2 equation, the labor and treatment coefficients are $\hat{\beta} = 0.869$ and $\hat{\delta} = -0.180$, and the treatment–labor interaction is $\hat{\theta} = 0.034$.

In Case 2, the labor elasticity is $\partial y/\partial l = \beta + \theta x$. Evaluating it at the sample mean of lagged digitalization gives 0.882. The direct effect is $\partial y/\partial x = \delta + \theta l$; evaluating it at mean log labor gives 0.217. Thus, the sample-average indirect and direct effects are 0.463 and 0.217, respectively, nearly identical to the additive decomposition. The raw coefficient δ is the direct effect at $l = 0$, which lies far outside the observed support of log labor and is therefore not the effect for a typical firm. In Case 1 there is no interaction, so the labor elasticity is constant at 0.885.

An interquartile treatment increase of 0.25 index points implies, under the primary Case-1 specification, a 5.65 percent direct output increase (95 percent interval 2.06 to 8.52), a 12.30 percent labor-mediated increase (9.43 to 15.97), and an 18.64 percent total increase (13.77 to 23.40). These are model-based transformations of the log-point coefficients, not experimental comparisons.

5.2.1 Omitting productivity from the labor equation

To mirror the Monte Carlo misspecification, we re-estimate the labor equation after removing recovered productivity while retaining the same firms, firm-years, log-capital control, year indicators, and ACF production estimates. Consequently, any change in the decomposition comes only from the estimated treatment effect on labor.

Table 5: Effect of omitting productivity from the empirical labor equation

Estimand	Productivity controlled	Productivity omitted	Omitted minus controlled
Panel A: Case 1, additive			
Treatment effect on labor a_1	0.524 (0.066)	0.564 (0.066)	0.040 [-0.005, 0.041]
Indirect effect IE_L	0.464 (0.058)	0.499 (0.060)	0.035 [-0.004, 0.036]
Total effect TE	0.684 (0.079)	0.719 (0.082)	0.035 [-0.004, 0.036]
Mediated share IE_L/TE	0.679 (0.074)	0.694 (0.073)	0.016 [-0.002, 0.016]
Panel B: Case 2, interaction			
Treatment effect on labor a_1	0.525 (0.066)	0.564 (0.066)	0.039 [-0.006, 0.040]
Indirect effect IE_L	0.463 (0.058)	0.497 (0.060)	0.035 [-0.005, 0.036]
Total effect TE	0.680 (0.079)	0.715 (0.081)	0.035 [-0.005, 0.036]
Mediated share IE_L/TE	0.680 (0.074)	0.696 (0.073)	0.015 [-0.002, 0.016]

Notes. Both columns use the same sample and ACF production estimates. The controlled labor equation includes lagged digitalization, recovered productivity, log capital, and year indicators. The omitted equation removes only recovered productivity. Parentheses contain firm-block bootstrap standard errors. The final column reports the point difference and its paired bootstrap percentile interval (999 successful draws per case).

In Case 1, omitting productivity raises $\hat{a}_1$ from 0.524 to 0.564 and the indirect effect from 0.464 to 0.499; the total effect rises from 0.684 to 0.719. Case 2 gives nearly the same contrast. The direction agrees with the upward omitted-variable bias in the simulation. However, the paired bootstrap intervals for the empirical differences include zero. The empirical exercise therefore demonstrates sensitivity to the productivity control, whereas the Monte Carlo experiment—where the truth is known—establishes the bias.

The numerical moments are solved accurately. Under the raw, uncentered Case-2 parameterization, however, the minimum Jacobian singular value is 3.14×10^{-4} and the condition number is 3.09×10^{3}. Because the instrument columns but not the parameter columns are standardized, these diagnostics are sensitive to coefficient scale and are not invariant weak-identification tests. In particular, the raw treatment coefficient is defined at $l = 0$ and is imprecisely estimated, with a 95 percent interval of $[-1.305, 0.831]$. A draw-by-draw transformation to average marginal effects agrees with the centered calculation to within 1.2×10^{-7}, confirming that the reparameterization does not change the reported average direct, indirect, or total effects. The proxy first-stage R^2 is 0.848 and the fitted materials derivative is positive for 92.5 percent of observations. Because this share is not one, the fitted sieve does not support global monotonicity over the whole empirical sample.

Table 6: Sample and ACF diagnostics for the labor-mediation application

	Case 1	Case 2
First-stage observations	17,277	17,277
ACF observations	13,902	13,902
Firms	2,416	2,416
Proxy first-stage R^2	0.848	0.848
Positive proxy derivative	0.925	0.925
Productivity-law R^2	0.839	0.838
Labor-equation R^2	0.777	0.776
Maximum absolute moment	8.68e-13	1.91e-12
Minimum Jacobian singular value	1.74e-02	3.14e-04
Jacobian condition number	2.53e+01	3.09e+03

Notes. Both structural systems are exactly identified. The proxy derivative is computed from the fitted year-specific sieve. The Jacobian diagnostics are calculated after instrument standardization but without parameter standardization, so they are sensitive to the coefficient scale used in the uncentered Case 2 specification.

6 Discussion

When labor is the mediator, the indirect effect has a direct production interpretation. Digitalization first changes labor hours by $\hat{a}_1$, and the resulting labor change affects output through the labor elasticity. The two specifications yield nearly the same average decomposition. In Case 1 (the baseline additive model), the estimated capital and labor elasticities are 0.200 and 0.885, respectively. The estimated direct effect of a full index-point increase in lagged digitalization is 0.220 log points. The controlled treatment coefficient in the labor equation is $\hat{a}_1 = 0.524$, which yields an indirect effect of 0.464 and a total effect of 0.684.

In Case 2, the uncentered labor coefficient is $\hat{\beta} = 0.869$ and the marginal labor elasticity is $\partial y/\partial l = \beta + \theta x$. Evaluating this expression at $x = \bar{x}$, the sample mean of lagged digitalization, gives 0.882. Likewise, the uncentered treatment coefficient is $\hat{\delta} = -0.180$, while the direct effect at mean log labor is $\delta + \theta\bar{l} = 0.217$. Case 2 estimates the treatment–labor interaction at $\theta = 0.034$. Because this interaction is small relative to its uncertainty, allowing the labor elasticity to vary with treatment changes the average direct and indirect effects very little. Therefore, the additive model adequately summarizes the average relationship in this application. To contextualize the economic magnitude under the primary Case-1 specification, an interquartile treatment increase of 0.25 index points implies a 5.65 percent direct output increase (95 percent interval 2.06 to 8.52), a 12.30 percent labor-mediated increase (9.43 to 15.97), and an 18.64 percent total increase (13.77 to 23.40). These are model-based transformations of the log-point coefficients, not experimental comparisons.

To empirically mirror the Monte Carlo misspecification, we re-estimate the labor equation after removing recovered productivity while retaining the same firms, firm-years, log-capital control, year indicators, and ACF production estimates. Consequently, any change in the decomposition is driven solely by the estimated treatment effect on labor. In Case 1, omitting productivity raises $\hat{a}_1$ from 0.524

to 0.564 and the indirect effect from 0.464 to 0.499; the total effect rises from 0.684 to 0.719. Case 2 yields nearly the same contrast. While this direction agrees with the upward omitted-variable bias established in the simulation, the paired bootstrap intervals for the empirical differences include zero. Thus, the empirical exercise demonstrates sensitivity to the productivity control, whereas the Monte Carlo experiment—where the true data-generating process is known—definitively establishes the bias.

The diagnostics therefore support numerical convergence while cautioning that the raw Case-2 level coefficients are less well conditioned than the average marginal effects. The fitted materials derivative remains positive for 92.5 percent of observations, so the sieve does not support global monotonicity over the entire empirical sample.

Finally, our causal interpretation rests on several structural and temporal assumptions. The ACF strategy requires materials demand to be strictly invertible in a scalar productivity state conditional on the other production variables. The fitted positive-derivative share serves as a diagnostic for that restriction, not a formal proof. Multiple productivity dimensions, adjustment costs in materials, input-price heterogeneity, or measurement error could invalidate this inversion. Furthermore, the labor equation estimates how lagged digitalization shifts labor conditional on recovered productivity, log capital, and year indicators. While the proxy controls for the scalar productivity state entering the production function, it does not automatically remove every omitted determinant of labor demand. Accordingly, the mediation interpretation continues to require the conditional exogeneity of lagged digitalization in the labor equation.

Lagging digitalization makes the treatment predetermined relative to the current productivity innovation, supporting its use in the ACF moment conditions. However, predetermination is weaker than random assignment. Lagged digitalization may remain correlated with persistent demand conditions, organizational changes, or anticipated hiring. The empirical decomposition is therefore causal only if these remaining pathways are sufficiently captured by the observed controls and the recovered productivity. While an external treatment instrument could replace this restriction, instrument-based mediation generally requires additional strong exclusion restrictions (Frölich and Huber, 2017). Future extensions could explore more flexible sieves, alternative proxies, and weak-identification-robust inference to further relax these constraints.

7 Summary and Conclusion

In this paper, we develop a structural framework for causal mediation analysis within production functions. Economists are frequently interested in decomposing the total effect of a treatment variable, such as digitalization, into a direct technology effect and an indirect effect operating through an endogenously chosen input, such as labor. Standard mediation techniques are invalid in this setting because firms choose both the treatment and the mediator conditional on latent productivity. To overcome this pervasive simultaneity problem, we propose a sequential estimator that combines a proxy-variable

production-function estimator (ACF) with a productivity-controlled mediator equation. By extracting Hicks-neutral productivity in the first stage and utilizing it as a control function in the second, our approach consistently identifies the direct, indirect, and total effects of the treatment.

Our Monte Carlo experiments validate the proposed estimator across both additive and treatment–mediator interaction specifications. The simulations demonstrate that the feasible ACF procedure accurately recovers the underlying structural parameters and the true mediated share. Crucially, the experiments highlight the danger of partial structural corrections. When the treatment and the mediator are both confounded by unobserved productivity, correcting only the production function is insufficient. We show that deliberately omitting the recovered productivity control from the mediator equation induces a substantial upward bias in the estimated treatment–labor coefficient, thereby severely overstating the indirect effect.

We apply this methodology to a panel of Spanish manufacturing firms to quantify the direct and labor-mediated effects of digitalization on value-added output. In our baseline additive model, we find that the indirect effect of digitalization operating through labor adjustment (0.464) is substantially larger than its direct productivity effect (0.220). In economic terms, an interquartile increase in the digitalization index implies a 5.65 percent direct output increase alongside a 12.30 percent labor-mediated increase. Furthermore, allowing the labor elasticity to vary flexibly with the treatment yields a small and statistically imprecise interaction parameter, suggesting that the additive specification adequately captures the average firm response.

Consistent with our simulation evidence, we demonstrate that estimating the empirical labor equation without the latent productivity control upward-biases the indirect effect from 0.464 to 0.499. While the paired bootstrap interval for this specific empirical difference includes zero, the direction of the shift confirms the model's sensitivity to omitted-productivity bias.

Ultimately, our framework provides applied researchers with a structurally grounded method to look inside the "black box" of the firm. By formally linking causal mediation with proxy-variable techniques, this approach enables a precise decomposition of how technologies, policies, or other firm-level treatments propagate through endogenous input choices to affect final output.

Declarations

Generative AI. We used Claude Code (Anthropic) and OpenAI Codex to assist with simulation and numerical checks. We reviewed all generated code and output, made all substantive methodological and interpretive decisions, and take full responsibility for the article. *Funding.* None. *Competing interests.* None declared. *Data availability.* The ESEE is administered by the SEPI Foundation on behalf of Spain's Ministry of Industry; the microdata are available to researchers on application and are not publicly redistributable.

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A Variable Definitions

- **Investment (I_{it}):** The value of current-period investment in equipment goods, excluding buildings, land, and financial assets. Investment made in period t becomes productive in period $t+1$.

- **Capital (K_{it}):** The real capital stock, computed using the perpetual inventory method as $K_{it} = (1-\delta)\frac{P_t^I}{P_{t-1}^I}K_{it-1}+I_{it-1}$, where $\delta = 0.15$ is the annual depreciation rate and P_t^I is the manufacturers' investment price index. Real capital is obtained by deflating capital at current replacement values by the investment price index.

- **Labor (L_{it}):** Total effective hours worked by all employees of the firm in year t.

- **Materials (M_{it}):** The real value of intermediate consumption, comprising raw materials, energy, and purchased services. Nominal intermediate consumption expenditures are deflated by a firm-specific intermediate consumption price index constructed from price variations reported in the ESEE.

- **Value Added Output (Y_{it}):** The real value of goods and services produced, obtained by deflating nominal production values by a firm-specific output price index constructed using the Paasche index formula less the real intermediate consumption.

- **Wage ($W_{L,it}$):** The hourly wage rate, defined as total labor compensation divided by total effective hours worked.

- **Output Price Index (P_{it}^Y):** A firm-specific Paasche price index constructed from price variations of output sales reported in the ESEE, following the methodology of Jaumandreu and Lin (2018).

- **Materials Price Index (P_{it}^M):** A firm-specific Paasche price index constructed from price variations of intermediate consumption reported in the ESEE.

B Digitalization Index Components

Table 7 details the thirteen variables that comprise the digitalization index utilized in the empirical application.

Table 7: Digitalization Index: Variable Definitions

Dimension	**Variable**	**Definition**
Automation	Use of robots	Whether the company uses robots in the production process (Yes/No)
	Use of CNC machine tools	Whether the company uses computer numerically controlled machine tools in the production process (Yes/No)
	Use of computer-aided design	Whether the company makes use of computer-assisted design (Yes/No)
	Use of flexible systems	Whether the company combines machine tools controlled by computer, robotics, and/or CAD through a mainframe computer (Yes/No)
	Use of LAN	Whether the company uses a local area network in its manufacturing activities (Yes/No)
Interactions with stakeholders	Ownership of internet domain	Whether the company has its own internet domain (Yes/No)
	Ownership of a webpage	Whether the company's web page is stored on its own servers (Yes/No/NA)
	Sales to companies via internet	Availability of a sales system for other companies through internet (Yes/No/NA)
	Sales to final consumer via internet	Availability of a sales system for final customers through internet (Yes/No/NA)
	Purchases from suppliers online	Whether goods or services are purchased through the internet (Yes/No/NA)
Digital-related human capital	External expenditure on IT training	Value of external expenditure on employee training in computing and information technologies (Euros)
IT technological component	Investment in computer equipment	Share of purchases of information processing equipment in total tangible fixed asset purchases (%)
	Hiring of software programming	Whether the company contracts out software-programming services (Yes/No)

Notes: The global index contains the 13 variables shown. ESEE adoption codes are explicitly recoded as 1 for Yes and 0 for No; non-response codes are treated as missing. Investment in computer equipment is converted from a percentage of total capital-goods investment into an absolute flow before capitalization. Computer-equipment investment and IT-training expenditure are capitalized with 15% depreciation. Quadrennial indicators are carried forward within firms for at most three years.